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Implement Simple Linear Regression using Scikit-Learn.

Code:

```python
# Importing libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Initializing Data
X = np.array([5,15,25,35,45,55]).reshape((-1,1))
y = np.array([5,20,14,32,22,38])

# Creating Fit Model
model = LinearRegression().fit(X,y)
# fit() = best fit straight line for data

# View Results
print("Intercept:", model.intercept_)
print("Slope:",model.coef_)
# y change for each unit change in X
print("R² score:", model.score(X,y))
# R² = variance in data
# R=1 -> Perfect, R=0 -> No linear relation

# Predictions
y_pred = model.predict(X)
print("Predictions:", y_pred)
```

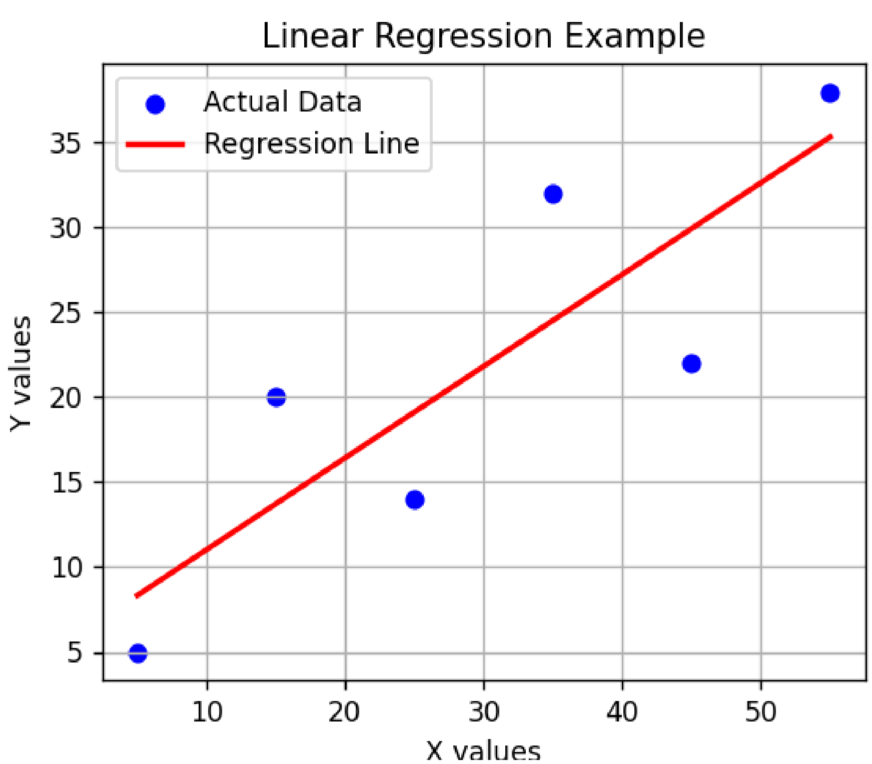
```python
# Plot
plt.figure(figsize=(5,2))
plt.scatter(X, y, color='blue',
            label = 'Actual Data')
plt.plot(X, y_pred, color = 'red',
         linewidth = 2, label = 'Regression Line')

plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Linear Regression Example")

plt.legend()
plt.grid(True)
plt.show()
```

Output:

```
Intercept: 5.633333333333329
Slope: [0.54]
R² score: 0.7158756137479542
Predictions: [ 8.33333333 13.73333333 19.13333333 24.53333333 29.93333333 35.33333333]
```



ssImplement Logistic Regression using Scikit-Learn.

```python
import numpy
from sklearn.linear_model import LogisticRegression

X = numpy.array([3.78,2.44,2.09,0.14,1.72,1.65,
                 4.92,4.37,4.96,4.52,3.69,5.88]).reshape(-1,1)
y = numpy.array([0,0,0,0,0,0,
                 1,1,1,1,1,1])

logr = LogisticRegression()
logr.fit(X,y)

predicted = logr.predict(numpy.array([3.46]).reshape(-1,1))
print(predicted)
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> & "C:\Users\Swaham Patra\AppData\Local\Programs\Python
[0]
```